(DCEL)

don't like that name but

2-cells ("faces") ← topological polygons

ptr to one edge defining it.

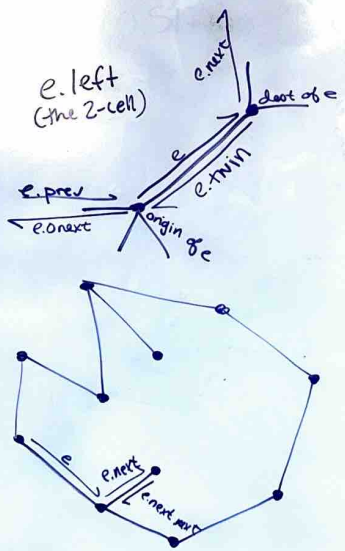
cell)

↑)

prev.twin

cell on e's "side")

n) as needed)



Questions

① How do we find a) destination vertex of e
b) all edges out from a vertex v
c) all vertices defining a 2-cell

② If we have 2 subdivisions (i.e., mfd rep'd by DCEL) of $I \times I \subset \mathbb{R}^2$, how can we update the two DCEL structures to create one "bigger" DCEL? (assume no verts shared)

③ What fails for non-orientability?

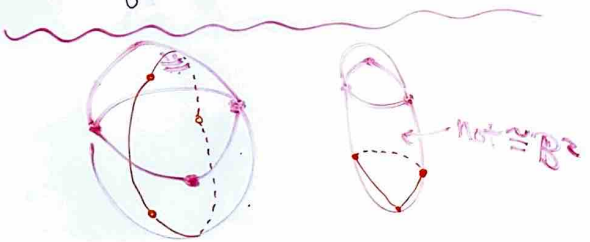
④ Can you think of a DS that will allow us to classify a compact 2-mfd (rep. as a CW-complex where attaching maps are homeomorphisms)?

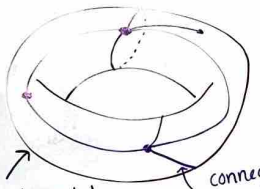
↑ orientable or not orientable?

Clarification

CW complex attaching map
$C \cong B^k$ (k-dim cell)
$\alpha : \partial C \to X_{K-1}$
K-1 skeleton

requirement: α is continuous.





orientable 2-complex T

connected graph such that T − G has cells all homeo to B^2

Q $K_5, K_{3,3}$ do not embed in $\mathbb{R}^2$, nor S^2
But, what about in $\mathbb{T}$? $\mathbb{RP}^2$? in K?

Data structure:
Doubly Connected Edge List (DCEL)

Vertices
ptr to one edge out of it

Edges
e.twin
e.origin
e.next (going ccw around 2-cell)
e.prev (opposite of ↑)
e.Onext = e.prev.twin
e.left (the 2-cell on e's "side")

2-cells ("faces") (don't like that name but topological polygons)
ptr to one edge defining it.

(can add more (or less) as needed)

